

EDUCATION:

B.S. Electrical Engineering, B.A. Physics
San Diego State University

GPA: 3.98
Expected Graduation: May 2028

WORK EXPERIENCE:

Electrical & Computer Engineering Intern, *VLI-EV Partners*

May 2025 - September 2025

- Conducted high-voltage testing and troubleshooting on a 115 MW solar farm battery energy storage system.
- Optimized and debugged a C++ BMS program running on a Linux operating system per customer requirements.
- Validated BMS software behavior through simulation and field testing to ensure accurate state-of-health estimation and system monitoring, and documented results for QC reports.
- Programmed and tested programmable logic controllers (PLC) to integrate Modbus-based safety alerts for ground faults and current leakage.
- Designed an AWS-hosted architecture and Python data analytics user interface to store, analyze, and visualize battery data for up to 25 years of future warranty lifetime.

Instructional Student Assistant, *SDSU Math and Science Learning Center*

August 2025 - Present

- Delivered one-on-one, drop-in tutoring in mathematics, physics, and computer science for undergraduates.
- Guided students through problem-solving using algebraic, graphical, and conceptual representations to promote independent learning.
- Graded homework assignments for a 70-student section of linear algebra.

RELEVANT EXPERIENCE:

Controls Lead Engineer, *Aztec Electric Racing*

September 2024 - Present

- Led a team of 10 software engineers responsible for vehicle software and diagnostic systems, including a Battery Management System (BMS), analog sensor integration, data acquisition, and CAN-based telemetry systems.
- Co-managed a \$13k annual controls budget to prioritize critical safety and rule-compliant upgrades for Formula SAE EV competition.
- Redesigned CAN bus and sensor wiring harness using electromagnetic compatibility best practices to restore real-time telemetry and improve signal reliability.
- Refactored BMS software for migration to STM32 microcontroller hardware to improve finite-state-machine control logic and CAN communication with the inverter control system.
- Documented software architecture, diagnostic workflows, and high-voltage vehicle test procedures for internal use and competition design presentations.

Undergraduate Researcher, *Society of Physics Students*

October 2024 - Present

- Simulated ultrasonic interference with open-source software Ultraino and experimentally validated results against our student-built Schlieren imaging system, resulting in acoustic levitation of macroscopic particles.
- Validated PCB shields in Altium Designer; miniaturized electronics system and designed 3D printed enclosure.
- Developed and tested ultrasonic transducer phased array with FPGA-based PWM control in Verilog.
- Wrote grants, budget justifications, and research papers.

CERTIFICATIONS:

OSHA 10, CAL-OSHA, NFPA 70E, CRLA International Tutor Training Program level 1

SKILLS:

- Programming — C, C++, C#, Python, UNIX systems, LaTeX, Verilog, MATLAB, MIPS Assembly, Terraform.
- Design Tools — SPICE, Simulink, EasyEDA, KiCad, Altium Designer, Onshape.
- Lab Equipment — Analog & digital oscilloscopes, function generator, digital multimeter, NI LabView.
- Soft Skills — Technical writing, presentation, adaptability, time management, creative problem-solving.

PUBLICATIONS:

Co-author, “Schlieren Imaging of Ultrasound Wave Interference,” AIP Journal of Undergraduate Research in Physics and Astronomy (2025).